

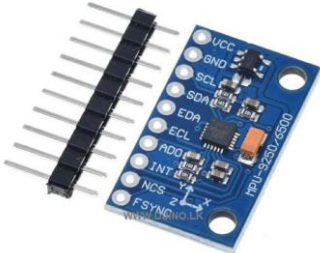
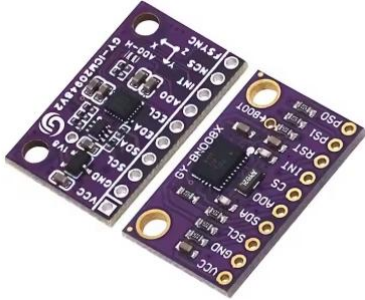
Interfacing Mobile Phone Inertial Measurement Unit (IMU) sensor data with MATLAB and Python

EE 2120: ELECTRICAL MEASUREMENTS AND INSTRUMENTATION

SEMESTER 4



Interfacing Mobile Phone IMU with MATLAB or Python via TCP or UDP

Inertial Measurement Unit (IMU)	Attitude and Heading Reference System (AHRS)	Inertial Navigation System (INS)
- Raw acceleration and angular rate data from Accelerometers + Gyroscopes (sometimes Magnetometer) - Output: Sensor Data Only  MPU 9250 9 DOF <i>Accelero+Gyro+Magnetometer Module</i>	- IMU sensor + Magnetometer + Fusion Algorithms - Outputs: Orientation(Roll, Pitch, Yaw)  High precision 9DOF AHRS sensor module	- Estimates Full Navigation state - Further integrates acceleration and orientation to estimate position and velocity



Interfacing Mobile Phone IMU with MATLAB or Python via TCP or UDP

- Install Hyper IMU on Mobile Phone.
- Connect the mobile phone with the PC via either Wi-Fi or USB tethering.
 - A. Wi-Fi come with delay, interference and packet loss.
 - B. USB tethering needs a wired connection.
- On the PC go to command prompt and run the command, *ipconfig* and identify the IPV4 address of the corresponding LAN Adapter.
- On the Hyper IMU app
 - Protocol: Either TCP or UDP
 - Server IP Address: IPV4 Address from previous step.
 - Server Port Number: 5555
 - Select the required sensors and Run
- In MATLAB or Python you can receive sensor data as either datagrams or bytes.
 - Port = 5555

The screenshot displays the HyperIMU mobile application interface, divided into two main panels. The left panel, titled 'HYPERIMU', contains settings for the connection. Under the 'Connection' section, the Protocol is set to 'UDP', the Sampling rate (ms) is '50', the Server IP address is '192.168.63.191', and the Server Port number is '5555'. The 'Persistent' option, which attempts to reconnect if the server is offline, is checked. The right panel, titled 'SENSORS LIST', shows a list of available sensors with toggle switches to select them. The selected sensors are LSM6DSL Accelerometer, MXG4300S Magnetometer, and LSM6DSL Gyroscope. Other sensors like STK33512 Light, STK33512 Proximity, MXG4300S Magnetometer Uncalibr., LSM6DSL Gyroscope Uncalibrated, Significant Motion, Step Detector, Step Counter, Tilt Detector, Pick Up Gesture, Device Orientation, Interrupt Gyroscope, Scontext, STK33512 Light CCT, and Call Gesture are currently disabled. A green checkmark icon is visible at the bottom right of the sensors list.

HYPERIMU	
Connection	
Protocol	UDP
Sampling rate (ms)	50
Server IP address	192.168.63.191
Server Port number	5555
Persistent	☒ Try to reconnect even if the Server is offline
Packet Configuration	
Header	Configure packet's header
Trailer	Configure packet's trailer
Prequel Packet	Transmit sensors names as first packet ☐
#-Terminator	Packets end with '#' instead of <CR><LF> ☐

SENSORS LIST	
☒	LSM6DSL Accelerometer
☒	MXG4300S Magnetometer
☒	LSM6DSL Gyroscope
☐	STK33512 Light
☐	STK33512 Proximity
☐	MXG4300S Magnetometer Uncalibr.
☐	LSM6DSL Gyroscope Uncalibrated
☐	Significant Motion
☐	Step Detector
☐	Step Counter
☐	Tilt Detector
☐	Pick Up Gesture
☐	Device Orientation
☐	Interrupt Gyroscope
☐	Scontext
☐	STK33512 Light CCT
☐	Call Gesture



Installing and Configuring Hyper IMU

